Problem 112P

A boat has an initial speed of $v_0 = 16$ ft/s. If it then increases its speed along a circular path of radius $\rho = 80$ ft at a rate of

$$\dot{v} = (1.5\,s)\ \text{ft/s}^2,$$

where s is measured in feet, determine the time required for the boat to travel a distance of $s = 50$ ft.

Step-by-Step Solution

Step 1

We are given that:

$$v_0 = 16\ \text{ft/s}, \qquad \rho = 80\ \text{ft}, \qquad \dot{v} = (1.5\,s)\ \text{ft/s}^2, \qquad s = 50\ \text{ft}.$$

We are asked to calculate the time required for the boat to travel the given distance.

Step 2

To calculate the speed of the boat, we use the relation:

$$v\,dv = \dot{v}\,ds$$

Step 3

Substitute the known values and integrate with appropriate limits:

$$\int_{v_0}^{v} v\,dv = \int_0^s (1.5\,s)\,ds$$

$$\left[\frac{v^2}{2}\right]_{v_0}^{v} = 1.5\left[\frac{s^2}{2}\right]_0^s$$

$$\frac{1}{2}(v^2 - v_0^2) = \frac{1.5}{2}s^2$$

$$v^2 = v_0^2 + 1.5\,s^2$$

Step 4

Substitute the numerical values:

$$v^2 = (16)^2 + 1.5\,s^2$$

$$v^2 = 256 + 1.5\,s^2$$

$$v = \sqrt{1.5\,(s^2 + 170.7)}$$

$$v = 1.2247\sqrt{s^2 + 170.7}$$

Step 5

To calculate the displacement, we use:

$$ds = v\,dt$$

Step 6

Substitute and integrate:

$$\int_0^s \frac{ds}{\sqrt{s^2 + 170.7}} = 1.2247 \int_0^t dt$$

$$\ln\left[s + \sqrt{s^2 + 170.7}\right]_0^s = 1.2247\,t$$

Step 7

Apply limits:

$$\ln\left(s + \sqrt{s^2 + 170.7}\right) - \ln(13.07) = 1.2247\,t$$

$$\ln\left(s + \sqrt{s^2 + 170.7}\right) - 2.57 = 1.2247\,t$$

$$t = \frac{\ln\left(s + \sqrt{s^2 + 170.7}\right) - 2.57}{1.2247}$$

Step 8

Substitute $s = 50$ ft:

$$t = \frac{\ln\left(50 + \sqrt{50^2 + 170.7}\right) - 2.57}{1.2247}$$

$$t = \frac{\ln(101.68) - 2.57}{1.2247}$$

$$t = \frac{2.05}{1.2247}$$

$$\boxed{t = 1.67 \text{ s}}$$